
PROMPT INJECTION DEFENSES FINAL REPORT *

Adrian Lopez
Department of Computer Science
Texas A&M University
San Antonio
Alope0321@jaguar.tamu.edu

Mallory Sorola
Department of Computer Science
Texas A&M University
San Antonio
Msoro02@jaguar.tamu.edu

ABSTRACT

Prompt injection is a growing security risk for applications that use large language models (LLMs), especially systems that combine trusted instructions with untrusted user or retrieved text. To prevent Prompt Injection's many companies implement tools like Prompt Injection Scanners that will detect malicious content like this.

Although a great tool, it is not fully reliable when used alone. So we conducted a case study evaluating similar tools, where we observed distinct differences in Rebuff, Vigil, and Prompt Injection. Rebuff relies on a heuristic scoring mechanism that proves effective at flagging both obvious and borderline threats by evaluating risk against a defined threshold. In contrast, Vigil utilizes a YARA-based framework that offers high transparency and precision for established attack signatures, though it struggled with subtle inputs that fell outside its predefined rule set. Along with that, Prompt Injection focuses solely on detecting and filtering malicious input. Comparing these tools revealed that, when used alone, they can be considered weak, but working together they form a stronger prompt injection tool.

Thus, we developed a multi-layer detection pipeline that integrates a heuristic layer inspired by Rebuff with the machine learning classifier from PromptInjection. Our experimental results indicate that this ensemble approach significantly bolsters detection robustness by compensating for the specific limitations of each individual method. For instance, while the heuristic component efficiently identifies explicit keyword-based threats, it often overlooks nuanced prompts, that lack obvious indicators. The PromptInjection model, however, successfully flagged these instances by analyzing underlying intent rather than just surface patterns. By combining these two strategies, the integrated system achieves a broader detection coverage than any single tool, meeting our objective of enhancing and unifying existing frameworks into a more robust defensive solution.

Keywords Prompt Injection Defenses · Rebuff Tool · Vigil Tool

1 Introduction

Large Language Models (LLMs) are a common use in many modern applications today, while being widely used this can often make LLMs a target of malicious tools, like Prompt Injection. Prompt Injection is a malicious method meant to trick LLMs with malicious inputs that can alter machine instructions and behavior. Although there are many defensive tools to counter this type of injection attack, many of them offer a certain mitigation that can be taken advantage of and bypassed. To address these limitations, we conducted an evaluation of the Rebuff, Vigil, and Prompt Injection Tools. During this evaluation, we highlighted their complimentary strengths and weaknesses in order to create a stronger tool. Thus, we propose a multi-layer detection pipeline that combines heuristic analysis from the rebuff tool with machine learning classification from the prompt injection tool to overall improve coverage and robustness.

**Citation:* Authors. Title. Pages.... DOI:000000/11111.

2 Background & Related Work

3 System Description

4 Proposed Contributions

5 Experimental Setup

6 Results & Analysis

7 Limitations & Security Considerations

8 Conclusion & Future Work

References